The Riccati Vorticity Estimate $M' \le C - kM^2$ Fails for 3D Navier–Stokes on $\mathbb{T}^3$: An Explicit Counterexample and a Lean-Checked Audit of the Symplectic-Tether Draft

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Abstract

An earlier unpublished draft by the author claimed global regularity for the 3D incompressible Navier–Stokes equations via a "symplectic tether". Its central step was an a-priori Riccati estimate $M'(t) \le C - kM(t)^2$ for the vorticity sup-norm $M(t) = \|\omega(\cdot,t)\|_{L^\infty}$. We show that this estimate is false on $\mathbb{T}^3$ when C, k depend only on the viscosity ν. We do this with an explicit trigonometric initial datum whose peak vorticity strictly increases at $t = 0$. We also show it is false when C, k may depend in an arbitrary way on ν and the kinetic energy, or on ν and $\|u_0\|_{L^3}$, by a scaled localized vortex. If the constants may depend on the initial datum in an arbitrary way, the estimate is equivalent to global regularity and remains *open*. **This note makes no claim about the Clay Millennium Problem.** A companion Lean 4 audit of the draft proves that its tether form is symmetric and not antisymmetric (hence not symplectic), and that because the tether vanishes against the energy, $B(F, H) = 0$, it cannot change the flow. Any vorticity control therefore cannot come from the tether. The audit also proves that the draft's tanh solution formula cannot match initial data above $\sqrt{C/k}$. It proves the conditional chain "Riccati estimate + BKM $\Rightarrow$ no finite-time blow-up", with the estimate carried as an explicit hypothesis. Everything machine-checked builds without `sorry` and uses only Lean's standard axioms.

1 Setting and the estimate

Let $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, $\nu > 0$, and let u be a smooth solution of $\partial_t u + (u \cdot \nabla)u + \nabla p = \nu\Delta u$, $\nabla \cdot u = 0$, with vorticity $\omega = \nabla \times u$ satisfying

$$\partial_t \omega = (\omega \cdot \nabla)u - (u \cdot \nabla)\omega + \nu\Delta\omega. \tag{1}$$

Put $M(t) = \|\omega(\cdot,t)\|_{L^\infty}$. The draft's *key estimate* is

$$M'(t) \le C - k\,M(t)^2 \quad \text{on } [0,T), \qquad C \ge 0,\ k > 0. \tag{2}$$

Since M is in general only Lipschitz, we refute both the differentiable form (`DraftRiccatiEstimate`) and the weaker form with the upper right Dini derivative, $D^+M \le C - kM^2$ (`DiniRiccatiEstimate`). Any of these forms forces M to be non-increasing wherever $M > \sqrt{C/k}$.

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Variant	C, k may depend on	Status
V1	ν only (the draft's claim)	**false** (Theorem 3)
V2	ν and $\|u_0\|_{L^2}$, arbitrarily	**false** (Theorem 4)
V2′	ν and $\|u_0\|_{L^3}$	**false** (Remark 5)
V3	u_0 arbitrarily	equivalent to global regularity: **open**

Lemma 1 (touching lemma)**.** *(Lean: `rightDeriv_le_of_touch`, `draftRiccati_false_of_touch`, `diniRiccati_false_of_touch`.) Let $T > 0$ and $g \le M$ on $[0, T)$ with $g(0) = M(0)$, and let g have right derivative r at 0. If $r > C - kM(0)^2$, then neither form of (2) holds on $[0, T)$.*

2 V1: an explicit counterexample

Let

$$U(x, y, z) = 3(-\sin y,\ \sin x,\ 0) + (-\sin x \cos z,\ 0,\ \cos x \sin z), \qquad u_0 = A\,U,\ A > 0.$$

Proposition 2. *(Lean: `div_free`, `w1_eq`–`w3_eq`, `omega_sq_le`, `omega_at_origin`, `omega_norm_bound`, `rhs_origin`.)*

(a) $\nabla \cdot U = 0$ *and* $\Omega := \nabla \times U = (0,\ 2\sin x \sin z,\ 3(\cos x + \cos y))$.

(b) $|\Omega|^2 = 9(\cos x + \cos y)^2 + 4\sin^2 x \sin^2 z \le 36$, *with equality at the origin. So* $M(0) = \|A\Omega\|_\infty = 6A$.

(c) At $x = 0$ *the right-hand side of* (1) *for* $u = AU$, $\omega = A\Omega$ *equals* $(0, 0, 6A(A - \nu))$.

Proof sketch. (a) and (c) are direct differentiation; at the origin $U(0) = 0$, the strain is $S(0) = \mathrm{diag}(-1, 0, 1)$, $\Omega(0) = (0, 0, 6)$ and $\Delta\Omega(0) = (0, 0, -6)$. Stretching contributes $6A^2$, advection 0 and diffusion $-6\nu A$. For (b), $(\cos x + \cos y)^2 \le 2\cos^2 x + 2$ and $4\sin^2 x \sin^2 z \le 4(1 - \cos^2 x)$, so $|\Omega|^2 \le 22 + 14\cos^2 x \le 36$. □

Theorem 3 (V1 is false)**.** *For every $\nu > 0$, $C \ge 0$ and $k > 0$ there is $A > \nu$ (explicitly $A = \nu + 1 + |C|/k$) such that the smooth solution with $u_0 = AU$ violates* (2)*, in either form, on every interval $[0, T)$ with $T > 0$.*

Proof. By classical local well-posedness [2, 3], the solution is smooth on $[0, T) \times \mathbb{T}^3$ up to $t = 0$. Hence $g(t) := \omega_3(0, t)$ is C^1 with $g'(0) = 6A(A - \nu)$ by Proposition 2(c) and (1). Also $g \le |\omega(0, t)| \le M(t)$ and $g(0) = 6A = M(0)$. (Using the component ω_3 avoids differentiating $|\cdot|$.) The arithmetic lemma `amplitude_exists` gives $6A(A - \nu) > C - k(6A)^2$, and Lemma 1 concludes. □

Concretely, with $\nu = 1$ and $A = 3$ we get $M(0) = 18$, and the lower right Dini derivative of M at 0 is at least 36.

3 V2: constants depending on ν and the energy

Amplitude scaling alone cannot refute V2, since an arbitrary $C(E)$ can absorb the growth of the energy E. We use Navier–Stokes scaling of a localized field instead.

Base field. Let $\beta(\rho) = \exp(-3\rho/(1-\rho))$ for $\rho < 1$ and $\beta = 0$ for $\rho \geq 1$, with $\rho = |x|^2$. Let $\Psi = \beta\, e_z + z\beta\,(-y, x, 0)$ and $u_1 = \nabla \times \Psi$. This u_1 is smooth, divergence-free and supported in the closed unit ball. Exact symbolic computation (`scripts/v2_symbolic.py`) gives $\omega_1(0) = (0, 0, 12)$, $u_1(0) = 0$, $\partial_3 u_{1,3}(0) = 2$ and $\Delta\omega_1(0) = (0, 0, -120)$. It also shows that $|\omega_1|^2$ is exactly invariant under rotations about the z-axis and even in x and in z. A rigorous ball-arithmetic certificate (`scripts/v2_maxcert.py`, python-flint/Arb [7]) proves $\max_{\mathbb{R}^3} |\omega_1|^2 = 144$, attained only at the origin, so $\|\omega_1\|_\infty = 12$. The certificate has three parts: (I) negative-definite interval Hessian on $[-\frac{1}{8}, \frac{1}{8}]^2$ (592 boxes); (II) interval upper bounds < 144 on all other boxes meeting $\rho \leq \frac{19}{20}$ (4,345 evaluations); (III) an explicit tail bound $\leq 2.8 \cdot 10^{-35}$ for $\frac{19}{20} \leq \rho < 1$.

Family. Fix $B = 10\nu$. For $\lambda \geq 1$ put $\lambda B\, u_1(\lambda x)$ in the ball of radius $1/\lambda$ about $0 \in \mathbb{T}^3$, and add a background $\mu\, u_2(x - p)$ with $p = (\pi, \pi, \pi)$. Here $u_2 = \nabla \times (\beta(|x|^2)e_z)$ is a fixed nonzero smooth divergence-free field supported in the unit ball, with finite $M_2 = \|\nabla \times u_2\|_\infty$ and energy $E_2 > 0$. The two supports are disjoint. The energy is $\frac{1}{2}\|u_0\|_2^2 = \lambda^{-1}B^2E_1 + \mu^2E_2$. So for any target $E^* > 0$ and $\lambda > B^2E_1/E^*$, the choice $\mu = \sqrt{(E^* - \lambda^{-1}B^2E_1)/E_2}$ makes the energy exactly E^*. For λ large, $M(0) = \max(12B\lambda^2, \mu M_2) = 120\nu\lambda^2$, attained at 0. At $(x, t) = (0, 0)$ the background vanishes nearby, and

$$\partial_t\omega_3(0,0) = (12\lambda^2 B)(2\lambda^2 B) - 0 + \nu\lambda^4 B(-120) = 24B(B - 5\nu)\lambda^4 = 1200\,\nu^2\lambda^4.$$

Theorem 4 (V2 is false)**.** *For every $\nu > 0$, $E^* > 0$ and every choice $C = C(\nu, E^*) \geq 0$, $k = k(\nu, E^*) > 0$, some smooth divergence-free u_0 on $\mathbb{T}^3$ with energy exactly E^* violates* (2) *at $t = 0$.*

Proof. As in Theorem 3, with $g(t) = \omega_3(0, t)$, $g(0) = M(0) = 120\nu\lambda^2$ and $g'(0) = 1200\nu^2\lambda^4$. Choose λ with $1200\nu^2\lambda^4 \geq C$. Then `localized_family_violation` gives $C - k(120\nu\lambda^2)^2 < 1200\nu^2\lambda^4$, and Lemma 1 applies. □

Remark 5 (V2′). Without the background ($\mu = 0$), $\|\lambda B u_1(\lambda\cdot)\|_{L^3(\mathbb{T}^3)} = B\|u_1\|_{L^3(\mathbb{R}^3)}$ does not depend on λ. So constants depending on ν and the scale-critical norm $\|u_0\|_{L^3}$ are refuted by the same computation. Constants that may grow with a norm that blows up along the family, such as $\|u_0\|_{H^s}$ for $s > \frac{1}{2}$ or $\|\omega_0\|_\infty$, fall under V3.

4 V3 is equivalent to global regularity, and is open

For a fixed smooth u_0 on $\mathbb{T}^3$, the following are equivalent: (a) the solution is global and smooth; (b) there are $C \geq 0$ and $k > 0$ with $D^+M \leq C - kM^2$ for all $t \geq 0$.

(b)⇒(a): comparison gives $M \leq \max(M(0), \sqrt{C/k})$, so $\int_0^T M < \infty$ for every T, and the Beale–Kato–Majda criterion [1] excludes blow-up.

(a)⇒(b): the elementary Danskin-type bound $D^+M(t) \leq \|\partial_t\omega(t)\|_\infty$ holds. For a global smooth solution, eventual smallness of $\|\nabla u\|_2$, small-data theory and decay of higher norms [3, 4] (after a Galilean reduction to mean zero if $\int u_0 \neq 0$) give $\sup_t \|\partial_t\omega\|_\infty < \infty$ and $\sup_t M < \infty$.

Hence V3 for all smooth data is equivalent to global regularity on $\mathbb{T}^3$, Fefferman's statement (B) [6] up to the choice of period. *It is open, and nothing here proves or disproves it.*

At a point where $|\omega|$ is maximal, $\partial_t|\omega| = |\omega|\,\xi \cdot S\xi + \nu\,\xi \cdot \Delta\omega$ with $\xi = \omega/|\omega|$. The stretching term has size $\sim M^2$ and can be positive (its value was $+1$ in V1 and $+2$ in V2 per unit amplitude) [5]. Nothing in (1) produces a $-kM^2$ term.

5 Lean-checked audit of the symplectic-tether draft

In the draft, $B(F,G) = -\kappa W\langle P^{\perp}\delta F, P^{\perp}\delta G\rangle$, where $P^{\perp}w = w - \frac{\langle w,u\rangle}{\|u\|^2}u$ and $W = \int|\omega|^2$. The "tether 2-form" is $\mathfrak{T}(a,b) = \omega\cdot(a\times b) - \kappa|\omega|^2(a\cdot b)$. The file `lean/FrohmanianTetherDraftAudit.lean` proves the following.

- `tetherB_symm`, `tetherB_degenerate_H`: B is symmetric and $B(F,H)=0$ for $\delta H = u$.
- `tetherB_diag_ne_zero`, `tetherB_not_antisymm` (and a concrete $\mathbb{R}^2$ witness): if $\kappa, W \neq 0$ and $P^{\perp}v \neq 0$ then $B(v,v)\neq 0$, so B is *not* antisymmetric.
- `arnold_part_antisymm`, `tether2Form_not_antisymm`: the Arnold part is antisymmetric, but $\mathfrak{T}$ is not, so it is not a symplectic form, as the draft's A1, Lemma 3 and Theorem 4.1 had claimed.
- `generator_indep_kappa`, `isTetherFlow_iff`, `observable_deriv_indep_kappa`, `draftRiccati_tether_iff_classical`: because $B(F,H)=0$, the generated dynamics do not depend on κ. The key estimate holds for the tethered flow if and only if it holds for the classical flow, so the claimed feedback cannot come from the tether. (Here `IsTetherFlow` is an abstract weak formulation of the draft's bracket dynamics, not a PDE solution concept.)
- `tanh_solves_riccati`, `tanh_initial_lt`, `tanh_cannot_match_large_data`: the draft's formula $y = \sqrt{C/k}\tanh(\sqrt{Ck}\,t+\phi_0)$ solves $y' = C - ky^2$, but always has $|y(0)| < \sqrt{C/k}$. It can never match data above $\sqrt{C/k}$.
- `riccati_comparison`, `riccati_integral_bound`, `conditional_extension`, `conditional_no_finite_blowup`: the *conditional* chain. If `DraftRiccatiEstimate` and the BKM continuation property (`BKMContinuation`) hold, both as explicit hypotheses, then the solution extends past every finite time. This is only conditional; by Theorems 3–4, the hypothesis is false in the draft's form.

6 What is machine-checked and what is cited

In Lean 4 [8] with Mathlib [9] (both files, 50 declarations, no `sorry`, no `axiom` declarations):

- the touching lemma, in both differentiable and Dini forms;
- for $u_0 = AU$: the divergence, the curl, the global bound $|\Omega| \le 6$ with equality at 0, and the exact right-hand side of (1) at the origin (derivatives as `deriv` of coordinate slices);
- the amplitude choice; the scalar step of V2;
- all audit statements in Section 5.

Cited or computed outside Lean:

- Local existence, uniqueness and smoothness up to $t=0$ for smooth data on $\mathbb{T}^3$ [2, 3]. This identifies $g'(0)$ with the right-hand side of (1), the derivative step behind the touching lemma.
- The Danskin-type bound $D^+M \le \|\partial_t\omega\|_\infty$ (used only in the V3 remark).
- The BKM criterion [1], which is an explicit hypothesis in Lean.
- The V2 symbolic values (sympy) and the Arb maximum certificate [7], plus the energy and support bookkeeping of the family.

- The decay and eventual-smallness facts in Section 4 [3, 4].

Numerics are supporting only. A pseudo-spectral simulation (`scripts/ns_check.py`, optional in `verify.sh`) is not part of any proof. With $\nu = 1$, $A = 3$ on a 64^3 grid, it shows an initial slope of 35.80 against the exact 36. For the localized field with $B = 10$, $\lambda = 1$, the slope is 1162 at 192^3 against the exact 1200. The 128^3 run is under-resolved.

7 Reproducibility

Cite as: B. S. Frohman, *The Riccati Vorticity Estimate $M' \leq C - kM^2$ Fails for 3D Navier–Stokes on $\mathbb{T}^3$: An Explicit Counterexample and a Lean-Checked Audit of the Symplectic-Tether Draft*, version 1.0.2, Zenodo, 2026, doi:10.5281/zenodo.24102417. All versions: doi:10.5281/zenodo.24102220.

Repository:

`https://github.com/BenFrohman/ns-tether-key-estimate-refutation`

The Lean sources, scripts and `verify.sh` are those of commit `44da3ff51c3f6145978891516916b068767a6ff6`. Tag `v1.0.2` adds only this PDF and the recorded hashes. Toolchain: Lean 4.22.0 (commit `ba2cbbf09d4978f416e0ebd1fceeebc2c4138c05`); Mathlib `v4.22.0`, revision `79e94a093aff4a60fb1b1f92d9681e407124c2ca`; Python 3.13.5 with python-flint 0.9.0, sympy 1.14.0, mpmath 1.3.0 and numpy 2.5.4. One command:

```
git clone https://github.com/BenFrohman/ns-tether-key-estimate-refutation
cd ns-tether-key-estimate-refutation
./verify.sh               # add --numerics for the optional simulation
```

Output of `lake env lean PrintAxioms.lean`, verbatim:

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